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FOOD SAFETY AND SANITATION PRACTICES IN LOCAL BAKERIES IN BUENASCAR; BASIS FOR POLICY ENHANCEMENT

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CHAPTER 1

THE PROBLEM AND ITS SCOPE

1.1 Background of the study

Food safety and sanitation are public health issues of great concern worldwide, with immediate effects on human health, societal well-being, and economic stability. According to the World Health Organization (2024), over 600 million people, nearly one in ten individuals in the world, fall ill due to consuming contaminated food, and 420,000 die from it, including 125,000 children under the age of five. These figures show that foodborne diseases continue to be a major global challenge. Beyond their health impact, unsafe food handling contributes to significant economic losses, especially in low- and middle-income countries spend over 110 million USD in healthcare costs and lost productivity due to unsafe food. These highlight the importance of proper sanitation and food safety practices in all food establishments, including bakeries, which serve as common and accessible sources of daily food in many communities. In the Philippines, food safety is governed by national regulations such as the Sanitation Code of the Philippines (P.D. 856) and the Food Safety Act of 2013 (R.A. 10611). Among these laws, P.D. 856 serves as the primary legal basis of this study because it provides clear and detailed standards for sanitation, including provisions on food handling, personal hygiene, equipment cleanliness, pest control, waste management, and sanitary inspections. These requirements are directly relevant to bakeries, where ingredients are manually handled and food products are prepared daily. However, despite the presence of these laws, many small and medium food establishments continue to face challenges in maintaining full compliance due to limited resources, outdated facilities, and insufficient monitoring, which may result in lapses in sanitation and inconsistent food safety practices.

International studies show that similar issues are observed in other countries. Research in Ghana and Poland found that while many bakeries follow basic hygiene routines, weaknesses in waste disposal, limited use of HACCP, and inconsistent enforcement still pose risks to consumers (Lawal et al., 2022; Drozd, 2022). Studies in Nigeria and Ethiopia likewise reported that inadequate facilities, lack of training, and irregular inspections contribute to unsafe food handling (Fakayode et al., 2022; Tamiru et al., 2022). Muhammad & Galadima (2023) confirmed that poor sanitation and lack of clean water were the two leading causes of microbial contamination in bread. Similar concerns were identified in Italy, where organizational issues and non-compliance affected bakery operations (Currò et al., 2024). In contrast, bakeries implementing HACCP, ISO 22000, and structured cleaning programs showed improved food safety and product quality (Bukhovets et al., 2022; Sahani, 2024; Padmavathi & Praveen, 2023). Despite these developments, many studies emphasize that gaps remain between food handlers' knowledge and their actual practices, underscoring the need for continuous training and supervision (Nurlena et al., 2023).

Studies in the Philippines reflect similar challenges. Research involving bakeries, karinderyas, and food stalls reported inconsistencies in hygiene and sanitation due to limited space, outdated facilities, and a lack of regular monitoring (Cardona et al., 2024; Ann et al., 2023). Studies conducted in Batangas, Marawi, and Zamboanga del Norte documented microbial contamination and unsafe handling practices among small food operators and street vendors (Tolentino et al., 2024; Pontino, 2024; Solon, 2022). Similar issues were observed in restaurants, where inconsistent cleaning routines and irregular facility maintenance were noted (Sarmiento & Apritado, 2022; Escario, 2025). Consumer studies also show that customers value clean surroundings, proper handling, and visible compliance with sanitation rules (Rivas et al., 2023). Although food handlers in areas like Isabela, Ifugao, and Marikina City generally demonstrate good

awareness of food safety principles, their actual practices remain moderate due to limited training or outdated equipment (Lucas et al., 2025; Shane et al., 2025).

Despite the availability of studies on food safety across the country, there is no existing research that specifically examines the sanitation and food safety practices of bakeries in Buenavista, Nasipit, and Carmen in Agusan del Norte. These municipalities have several long-standing bakeries that serve their communities daily, yet their level of compliance with sanitation standards remains undocumented. Without localized research, the actual conditions of their hygiene practices, equipment handling, and food safety procedures remain unclear. This gap underscores the need for a focused assessment of bakery operations in these areas. Therefore, this study aims to assess the food safety and sanitation practices of registered bakeries in Buenavista, Nasipit, and Carmen. It examines how bakery characteristics such as years of operation, number of employees, and business location along with consumer demographics relate to their hygiene practices, sanitation procedures, and equipment handling. Identifying these conditions is essential to determining strengths, addressing weaknesses, and supporting improvements.

This research is significant for multiple stakeholders. Bakery owners and workers may use the findings to evaluate their current sanitation practices and strengthen compliance with national standards, particularly those outlined in the Sanitation Code of the Philippines (P.D. 856). Consumers may become more aware of sanitation conditions and make safer choices when purchasing baked products. Local government units and municipal health offices may utilize the results to enhance monitoring systems and design targeted training programs. Finally, the study may serve as a reference for future researchers exploring similar topics, contributing to broader efforts to promote food safety and protect public health.

1.2 Theoretical Framework

This study is anchored on the Health Belief Model (HBM) (Rosenstock, 1977) and the Protection Motivation Theory (PMT) (Rogers, 1975). Together, these theories explain how bakery employees implement food safety and sanitation practices and how consumers evaluate these practices in the municipalities of Buenavista, Nasipit, and Carmen. These frameworks provide a behavioral foundation that links employee motivation, sanitation performance, and consumer perception within local bakery operations.

The Health Belief Model (HBM) is used to explain the hygiene and sanitation practices of bakery employees. According to this model, a person is more likely to follow preventive measures when they believe they are at risk of a problem, when they recognize that the consequences are serious, when they understand the benefits of taking action, and when they see fewer obstacles that might prevent them from complying. In bakery operations, employees' decisions to follow sanitation procedures such as handwashing, food handling, equipment cleaning, and waste management depend on their awareness of food contamination risks and the possible effects of foodborne illness. Characteristics of the bakery, such as the number of employees, years of operation, and location, may influence the availability of sanitation facilities and the workload. These factors shape how employees view both the benefits and barriers related to proper hygiene. Employees who understand the health and business consequences of improper sanitation are more likely to observe safe food handling practices.

The Protection Motivation Theory (PMT) is used to explain how consumers assess the sanitation practices of bakeries. PMT states that individuals judge a situation by considering the possible threat and the effectiveness of available protective actions. Consumer characteristics such as age, sex, educational attainment, and employment status influence how they notice and interpret hygiene conditions. When consumers observe aspects of the bakery such as staff

grooming, food handling, cleanliness of the environment, and the sanitation of equipment, they form judgments about the safety of the food. Good sanitation practices increase consumer confidence and may encourage continued patronage. In contrast, visible lapses raise concerns about risk and may affect satisfaction and purchasing decisions.

By combining HBM and PMT, the study uses two viewpoints. Employee behavior affects the actual level of sanitation, while consumer perception affects how these practices are evaluated. Food safety in bakeries results from the interaction of employee practices, consumer reactions, and the conditions present in each establishment. Differences in bakery characteristics influence how sanitation measures are carried out, while differences in consumer characteristics influence how these measures are interpreted. This framework provides a complete explanation of the behavioral and perceptual factors that influence food safety in bakeries and supports the objectives of this study.

1.3 Review of Related Literature for Food Safety and Sanitation Practice

This section presents related literature and studies on food safety and sanitation practices in bakeries. It includes discussions on respondent profiles, bakery characteristics, and hygiene and sanitation procedures, as well as research on customer assessment and how consumers evaluate cleanliness and safety in bakery operations. These sources provide the foundation for understanding the variables examined in this study.

Age

Age influences how individuals understand and apply food safety practices because maturity, experience, and exposure to household responsibilities often develop over time. The slow process of learning and unlearning creates the risk of bringing diverse age groups with differing knowledge and responsibilities to the same culinary point. Yi & Choi (2025) conducted research in an area around Seoul and found that people aged 40 to 60 had much better knowledge

of food safety practices than younger generations. The older people were very conscious and careful about food storage, hygiene, and contamination prevention. In contrast, young ones lacked knowledge of issues of paramount importance, such as mold and freezing methods for safeguarding food, which are all the more necessary for preventing foodborne diseases.

Similarly, evidence from Baghdad supports the notion that age significantly affects compliance with food safety standards. Studies reported that individuals aged 18 to 29 demonstrated lower adherence to established food safety guidelines compared to middle-aged and older adults (Abu-Almaaly, 2025). This trend indicates that life experience and knowledge gained over time may greatly influence safer, more responsible food-handling behaviors. With age, people usually acquire specific characteristics that indicate an awareness of their responsibilities, which might not yet be the case for younger people.

In addition, research conducted in Bangladesh indicates that generational differences strongly influence food safety literacy. Adults possessed better understanding of nutrition, sanitation, and safe food practices, whereas teenagers displayed notable gaps in knowledge and behavior (Islam et al., 2022). These differences can be explained by the practical experience that older individuals accumulate over time, contrasted with younger individuals who often rely on informal or less reliable sources of information.

Moreover, demographic studies in Saudi Arabia have reinforced the importance of age in shaping food safety attitudes and behaviors. Nationwide surveys showed that variables such as age and education strongly affect people's opinions and practices related to food safety (Almutlaq et al., 2025). While older generations tend to demonstrate greater maturity and consistency in maintaining safe food practices, younger generations, despite having better access to digital and social media sources of information, may not always develop a deep understanding of proper procedures.

Overall, the studies discussed above demonstrate a clear and consistent pattern: age plays a critical role in food safety awareness, attitudes, and practices. Older individuals are more likely to adopt safe and responsible food-handling behaviors, likely due to accumulated experience and lessons learned from past errors. In contrast, younger people often rely on rapid, informal learning methods, which can result in inconsistent or unreliable practices. These findings underscore the need for age-specific educational programs and awareness campaigns to strengthen the food safety knowledge of younger generations, reduce knowledge gaps between age groups, and promote a culture of safe and hygienic food handling across the population.

Educational Attainment

Educational attainment plays a critical role in shaping individuals' food safety awareness, nutritional habits, and hygiene practices. Higher education generally provides greater exposure to information on nutrition, public health, and food safety regulations, which encourages more responsible food handling and consumption. While numerous studies demonstrate a positive relationship between education and safe food practices, the strength of this association can vary depending on socioeconomic and cultural contexts. Understanding this relationship is essential for designing effective interventions that promote food safety across different populations.

Research in various international contexts highlights the influence of education on food choices and hygiene practices. In London, communities with higher levels of education were found to consume larger quantities of fruits, vegetables, and fish, whereas less educated communities favored bread, potatoes, and high-fat foods (Fard et al., 2021). This indicates that education can shape dietary decisions both directly and indirectly through community norms. Similarly, studies in Romania revealed that individuals with higher education not only demonstrated better food safety knowledge but also adopted safer kitchen practices and positive attitudes toward hygiene

(Mihalache et al., 2021). These findings underscore the pivotal role of education in promoting both awareness and practical application of food safety measures.

Further evidence supports the positive impact of education on food safety in diverse cultural and socioeconomic settings. In India, individuals with higher educational backgrounds were more likely to read and understand food labels, particularly for edible oils and fats, highlighting the role of education in improving label literacy and informed food choices (Paliwal et al., 2025). In Bangladesh, food vendors with higher education exhibited greater knowledge of hygiene and food safety practices compared to their less educated counterparts, suggesting that even basic education can improve awareness of foodborne infections and preventive measures (Islam et al., 2024). Similar patterns were observed in Ethiopia, where street vendors with higher education and proper training were more likely to practice safe hygiene, especially when supported by continuous monitoring and access to sanitation facilities (Negassa et al., 2023). In Spain, young people with higher education demonstrated healthier diets, lower body mass indices, and fewer unhealthy habits, including smoking and excessive consumption of sugary drinks (Sandri et al., 2024). Collectively, these studies indicate that education not only increases knowledge but also influences behaviors that promote food safety and overall health.

Education also produces lasting improvements in professional and organizational food safety practices. In Malaysia, continuous food safety education for hospitality students increased awareness and encouraged consistent hygiene practices (Ahmad et al., 2024). Similarly, structured education for bakery and pastry students in Laguna, Philippines, improved both practical skills and employability, demonstrating that education contributes to competency and career readiness in the food industry (Santos & Aquino, 2023). Beyond vocational training, long-term educational interventions can transform organizational culture. A case study of a military food service revealed that personnel gradually progressed from merely following rules to embracing food safety as a

core value, supported by ongoing education and leadership (Zanin et al., 2021). In Indonesia, differences in educational levels among university food handlers significantly affected knowledge, attitudes, and hygiene practices, although continuous training remained necessary to reinforce proper behaviors (Firdani, 2022). These examples highlight that education, when combined with training and supervision, is crucial for consistent and effective food safety practices in both institutional and professional settings.

Taken together, these studies demonstrate a strong and consistent relationship between educational attainment and food safety awareness, attitudes, and practices. Individuals with higher education are more knowledgeable about hygiene measures, more likely to store food safely, and more consistent in their safe food handling. Educational interventions, whether through formal instruction, public campaigns, or vocational training, have proven effective in promoting food safety across diverse contexts. Strengthening education, particularly among food handlers and consumers in local communities, can improve public health outcomes, enhance compliance with food safety regulations, and contribute to the development of a safer, more responsible food system.

Sex

Sex is a significant factor that influences food safety knowledge, attitudes, and practices, as it shapes individuals' responsibilities, awareness, and access to resources related to food handling. Men and women often differ in how they acquire, apply, and maintain knowledge regarding hygiene and safe food practices. Studies conducted among fruit and vegetable handlers in Ethiopia demonstrated that education improved knowledge for both men and women, with a stronger effect observed among women. Although the overall correlation between knowledge and practice was not pronounced, female handlers demonstrated a slight positive correlation,

indicating they may respond more effectively to educational interventions and targeted campaigns (Gebre et al., 2023).

In low- and middle-income countries, food safety responsibilities often fall disproportionately on women, particularly in household and informal food production. Research highlights that women face challenges, including limited access to training, technology, safe infrastructure, and decision-making authority, which increases their vulnerability to foodborne risks (Milovanova et al., 2024). Similarly, women in Ethiopia's dairy value chain play an indispensable yet often overlooked role, and resource constraints further expose them to contamination hazards (Garsow et al., 2022). These conditions indicate the need to address structural inequalities to improve food safety outcomes.

Gender differences also influence compliance and performance in professional food handling environments. A study examining male food handlers in China found that although many followed good practices, gaps remained in handwashing, food storage, and container use. Factors such as income, work experience, and training were associated with better compliance, whereas older age was linked to lower adherence (Chen, 2025). This suggests that male food handlers may benefit from targeted, skills-based interventions to address specific weaknesses in food safety practices.

Women often take primary responsibility for food handling at home and in informal sectors, facing barriers such as limited education, restricted decision-making power, and insufficient policy support. Men, by contrast, may have greater access to formal training and resources in professional food service settings but do not always exhibit consistent hygiene practices. Recognizing these gender-based differences is essential for designing interventions that improve compliance, knowledge, and safe food handling across all groups. Understanding how sex influences food safety practices allows for the development of more effective and inclusive

strategies. Gender-sensitive training, improved infrastructure, and equitable access to resources can empower both men and women to adopt safer practices. Addressing these differences in policy and practice can strengthen the overall food safety system and contribute to better public health outcomes.

Business Location

The location of a food establishment affects sanitation practices because access to water, waste management, and infrastructure varies widely across communities. Studies have shown that the physical site of a business, including access to infrastructure such as clean water, waste disposal, and proper shelter, affects food handlers' ability to maintain hygienic operations (Moges et al., 2024). In the Philippines, Molina et al. (2021) found that water, sanitation, and hygiene conditions vary widely across local communities, with some barangays experiencing limited access to safe water and adequate sanitation structures. These disparities contribute to inconsistent hygiene practices among establishments located in underserved areas. Environmental and infrastructure-related challenges are also seen in other countries. Bhandari and Bhusal (2021) reported that street food vendors in Pokhara, Kaski, struggled to maintain sanitation because of poor access to clean water and proper waste management systems, indicating that water supply directly influences compliance with hygiene standards. Moreover, Desye et al. (2023) found that food safety knowledge, attitudes, and practices among street food vendors in low- and middle-income countries are largely shaped by contextual and environmental conditions, including access to sanitation facilities, education, and income levels. Vendors operating in poorly developed areas often face structural challenges that limit their ability to maintain hygienic standards. Supporting this, Ndhlovu and Tabit (2024) reported that street food vending sites in Marabastad, Pretoria, exhibited inadequate sanitation and low compliance with food safety requirements, mainly due to the lack of proper facilities and infrastructure in their business locations. Likewise, Chumo et al.

(2023) emphasized that informal or marginalized settlements experience greater health and sanitation risks because of insufficient basic services, weak institutional support, and unsafe environments. These studies collectively highlight that business location plays a critical role in influencing food safety practices, as establishments situated in well-regulated, adequately serviced, and properly equipped areas are more capable of implementing and sustaining effective hygiene and sanitation measures.

Number of Employees

The number of employees in a bakery plays an important role in ensuring food safety, product quality, and overall operational efficiency. A properly structured workforce allows tasks to be distributed effectively, provides better supervision, and supports consistent hygiene and contamination control throughout the production process. Research has shown that microbial safety and contamination prevention in bakeries depend not only on equipment and methods but also on the management of the production environment, which is influenced by the size of the workforce (Vermelho et al., 2024). A sufficient number of employees makes it easier to monitor critical points such as fermentation, ingredient handling, and storage, thereby reducing the risk of microbial contamination.

Workforce size also affects how well spoilage and contamination risks are managed. Factors such as storage temperature and mold formation during cooling are closely linked to staff availability (Chou et al., 2024). When there are too few employees, it becomes difficult to maintain cleanliness, inspect equipment, and ensure proper storage conditions. These gaps increase the likelihood of microbial growth and product spoilage. On the other hand, an adequate number of employees allows responsibilities to be shared efficiently, enables prompt responses to potential hazards, and supports consistent implementation of hygiene measures.

Employee performance and bakery efficiency are further influenced by the number of workers. Studies in South Korea found that employees who were satisfied with their work and committed to the organization performed better, which positively affected overall bakery operations and food safety (Kang, 2021). Similar patterns have been observed in the Philippines. Labao et al. (2023) reported that food handlers with greater capacity to manage their workload and maintain systematic hygiene routines demonstrated better compliance with food safety practices. Their study showed that when enough personnel are present to divide tasks effectively, hygiene performance improves, helping prevent lapses in food handling and sanitation.

The evidence indicates that workforce size is a key factor in maintaining bakery hygiene and product safety. A sufficient number of employees helps enforce sanitation protocols, manage storage and cooling processes properly, and respond quickly to contamination risks. In contrast, understaffing can lead to lapses in hygiene, extended exposure of food to unsafe conditions, and higher chances of bacterial growth. To address these challenges, bakeries must maintain adequate staffing levels while providing continuous training and proper organization of their workforce. Ensuring that employees are both sufficient in number and well-trained strengthens operational efficiency and serves as a critical safeguard for safe and high-quality bakery products.

Years of Operation

Years of operation influence food safety performance because businesses with more experience tend to develop established systems, routines, and compliance with sanitation standards.In most cases, long-standing companies have well-organized and well-documented systems for handling food, maintaining hygiene, and complying with regulations. The study of Al Banna et al. (2021) emphasized that longer work experience among meat handlers in Bangladesh significantly contributed to improved food safety knowledge and practices. Handlers with more than ten years of experience demonstrated higher awareness of hygiene protocols and better

implementation of safe handling behaviors than less experienced workers. These findings highlighted that experience, along with education and training, serves as a strong predictor of food safety compliance.

The researchers also noted that older food establishments, which had undergone repeated inspections and regulatory updates over the years, had a higher likelihood of meeting sanitary standards. These businesses developed stronger skills in managing compliance, addressing infrastructure issues, and sustaining operational consistency, all of which contribute to improved hygiene performance. Similar observations were reported in the Philippines. Armas and Moralde (2024) found that ambulant food vendors with longer years of operation developed more systematic routines, demonstrated stronger organizational practices, and were better equipped to meet sanitation requirements compared to newer vendors. Their results show that operational longevity strengthens resilience and adherence to food-safety expectations. Chen et al. (2025) further supported this argument in their study of Chinese catering venues, where workers with longer experience showed higher compliance with safety procedures such as timely handwashing, food quality inspections, and contamination prevention. This suggests that establishments operating for many years tend to have more trained employees and more established safety routines than newly opened businesses.

A related study in Almaty, Kazakhstan, reported that employment duration was positively associated with food safety knowledge, attitudes, and practices (Kulpiisova et al., 2025). Food handlers with longer work experience displayed better safety performance, indicating that continuous exposure to food-handling environments enhances familiarity with hygiene procedures and encourages responsible behavior. In the Philippine context, a study of restaurants in Laguna and nearby areas also noted that establishments with more years of operation generally had more stable sanitation practices and stronger operational systems, reflecting accumulated

experience in meeting industry standards and regulatory expectations (Streamlining Operations and Best Practices in the Restaurant Industry, 2023). However, the Kazakhstan study also noted that many handlers had less than one year of experience, which correlated with lower knowledge levels despite generally positive attitudes.

Together, these studies suggest that the number of years an establishment or its workers have operated directly influences their food safety performance. Operational longevity not only reflects business stability but also represents accumulated learning, routine formation, and repeated exposure to regulatory requirements. These factors collectively strengthen compliance with food safety protocols over time.

Food Safety Practices

Practices related to food safety, particularly in personal grooming and hygiene, play a critical role in ensuring the overall quality of food and protecting consumer health. Many studies have shown that while food handlers are generally aware of hygiene standards, these standards are not always properly implemented. For instance, research in India and Vietnam revealed that food sellers often washed their hands and wore clean clothes, which are basic hygiene habits, but failed to use hairnets or wash their hands after handling money due to inadequate facilities and weak enforcement (Agrawal et al., 2025; Huynh-Van et al., 2022). In the Philippines, many eating houses were believed to follow sanitation regulations; however, shortages of water, protective gear, and supervision limited their ability to comply fully (Dollentas & Habla, 2021). Similarly, street and home-based sellers maintained some hygiene measures, yet laboratory tests showed bacterial contamination, highlighting a gap between perceived compliance and actual practice (Limon, 2021; Origenes et al., 2022). Studies conducted in Marawi and Zamboanga del Norte also reported that although food handlers possessed adequate knowledge, they struggled to meet sanitation standards due to insufficient training and infrastructure (Caye, 2024; Solon et al., 2022).

The personal grooming and hygiene practices of food handlers strongly shape food safety within food establishments. Lapses such as improper handwashing, unclean uniforms, and poor personal cleanliness can introduce contaminants into food preparation. Sanitation practices such as routine cleaning of equipment, utensils, and work areas, as well as proper waste management are likewise essential in preventing environmental sources of contamination. Research indicates that both hygiene-related behaviors and sanitation procedures significantly influence the overall effectiveness of food safety practices, making them essential components for maintaining safe and contamination-free operations (Bihag et al., 2021).

Moreover, education, training, and strong management support have consistently been shown to enhance workers' hygiene practices and overall compliance with food safety standards. Studies conducted in different settings have confirmed that employees who receive proper guidance and ongoing training maintain higher levels of cleanliness and sanitation (Tyler et al., 2023; Siddiky et al., 2024; Rifat et al., 2022). Similarly, restaurant staff who participate in regular food safety programs demonstrate superior hygiene and sanitation performance (Labović et al., 2023; Ermeje et al., 2025). These findings indicate that although food handlers recognize the importance of hygiene, sustainable compliance with food safety practices requires sufficient resources, consistent supervision, and institutional commitment to continuous training.

Sanitation Practices

Food sanitation is one of the issues in food safety, ensuring it is contaminant-free and maintaining the health of consumers. The research by Aulia et al. (2024) showed that although the majority of food handlers had satisfactory hygiene knowledge, many still engaged in poor sanitation practices, indicating that knowing what to do does not necessarily translate into practice. Very similar findings were reported by Pratama, (2023) during their assessment of the port canteens at Kendari; there, good sanitary conditions were observed mainly when workers

were appropriately trained and the environment was clean and well-equipped, demonstrating that both knowledge and environment matter. Vi M. et al. (2025) also noted in the Philippines that students with a better understanding of hygiene and sanitation are healthier, highlighting the role of sanitation habits in well-being.

On the contrary, Zamudioa & Cabanisasa, (2024) assessed the facility for smoked fish production. They discovered that, even with a high level of hygiene awareness, some operational routines were not consistently applied. All these studies indicate that food sanitation relies on training, knowledge, and the right environment for practices to take place, keeping food prepared and served safely and cleanly.

Customer Assessment

Customer perceptions of food service quality are closely connected to the food safety practices observed in an establishment. Picardal (2019) explained that customers usually evaluate sanitation based on three major areas: food service and personnel hygiene, sanitation in facilities and safety procedures, and utensils and equipment handling. These aspects include visible behaviors such as staff cleanliness, proper handling of baked products, the condition of the surroundings, and the cleanliness of the equipment used in daily operations. These indicators serve as the basis of how customers form judgments regarding the overall safety and cleanliness of a bakery.

Research shows that customer perceptions are strongly influenced by these visible hygiene cues. Sarmiento and Apritado (2022) reported that personnel hygiene, such as clean uniforms, proper grooming, and hygienic food handling, significantly improves customer evaluations. They also found that facility sanitation, including clean dining and preparation areas, orderly surroundings, and well-maintained restrooms, helps build customer confidence in the establishment. Proper cleaning, maintenance, and use of utensils and equipment further affect

customer satisfaction, which demonstrates how these three components work together in shaping customer perceptions.

Studies from other contexts support these findings. Wei (2021) noted that staff hygiene, sanitation of facilities, and proper handling of utensils play important roles in influencing customer satisfaction and trust. Owusu et al. (2021) found that clean facilities and effective waste management encourage repeat patronage because customers associate cleanliness with safety. Nicolosi et al. (2023) observed that health-conscious consumers tend to prefer bakeries that maintain visible hygiene practices. Nakpathom (2023) also reported that sanitation conditions, food hygiene standards, and the overall ambiance of an establishment influence customer loyalty.

The literature collectively shows that customers form their assessment of food safety and sanitation based on what they can clearly see. These indicators include the cleanliness of the bakery's surroundings, the hygienic appearance of staff, the organization of display counters, the appearance of products, and the absence of unpleasant odors or waste. Customer evaluation therefore extends beyond product quality and includes the establishment's visible level of hygiene, sanitation, and safety compliance.

Food Service & Personnel's Hygiene

Hygiene among food service personnel is essential for ensuring food safety and preventing foodborne illnesses, yet gaps in knowledge and actual practice remain widespread. Although many food handlers are aware of hygiene standards, proper implementation is often inconsistent because of limited training, insufficient resources, or weak supervision (Ibrahim & Yusuf, 2023). In Kano, Nigeria, only 18.7 percent of street vendors and 26.0 percent of canteen handlers demonstrated good hygiene knowledge. Common lapses such as inadequate handwashing and irregular use of protective gear show the need for regular training, certification, and stronger regulatory monitoring.

Even when food handlers possess good knowledge and positive attitudes toward hygiene, their practices may still be inconsistent. In Ilorin, 62.5 percent of fish vendors reported favorable attitudes toward food safety, especially in avoiding cross-contamination and maintaining personal hygiene. However, many did not consistently wear gloves or masks, which are necessary for preventing contamination (Ghali-Mohammed et al., 2024). Factors such as age and household responsibilities also influenced hygiene behaviors, suggesting that training programs and supervision need to consider demographic differences among workers.

Compliance with hygiene protocols is also shaped by workplace culture and operational limitations. In hospital food service facilities in Limpopo Province, South Africa, staff members understood hygiene requirements but were unable to follow them consistently because of cultural practices like wearing jewelry and operational challenges such as staff shortages, lack of proper uniforms, and insufficient inspection systems (Mphasha et al., 2024). These conditions hindered proper adherence to basic hygiene procedures.

Customer confidence in food establishments is closely linked to the visible hygiene practices of food handlers. Proper grooming, clean uniforms, regular handwashing, the use of protective gear, and hygienic handling of food strongly influence consumer perceptions of cleanliness and safety. The reviewed studies collectively show that improving personnel hygiene requires more than knowledge. It requires continuous training, supportive workplace conditions, adequate supplies, and consistent enforcement of hygiene standards to ensure safe food-handling behaviors.

Sanitation in Facilities & Safety Procedures

Sanitation in facilities and proper safety procedures are essential in foodservice operations because they help maintain food quality and prevent contamination. The study of Nurlena et al. (2023) assessed the sanitation and hygiene practices of 50 pastry production facilities and focused

on equipment cleanliness, water quality, and compliance with sanitation regulations. Although 68 percent of the establishments met minimum sanitation requirements, several remained weak in areas such as waste disposal, equipment cleaning, and staff hygiene training. The authors emphasized that continuous sanitation monitoring and regular staff training are necessary to ensure the safety and cleanliness of food facilities.

Differences in sanitation performance across countries also highlight the importance of consistent standards. Research by Na et al. (2024) comparing shared kitchens in Korea, Japan, and the United States revealed variations in sanitation guidelines, inspection systems, and documentation practices. Facilities with insufficient policies and limited training showed poorer hygiene outcomes. The study suggested that adopting uniform regulations, improving inspection processes, and strengthening training can enhance safety in food preparation areas.

Access to sanitation facilities also influences compliance with hygiene procedures. An investigation of 250 street food vendors in Pretoria found that those with limited sanitation resources and lower food safety knowledge were less likely to follow hygiene guidelines. Ndhlovu and Tabit (2024) explained that knowledge and access to facilities must be addressed together because improving one factor without the other leads to minimal impact on overall food safety.

Consistent with these insights, this study gives importance to visible aspects of facility sanitation that customers can directly observe. Clean display areas, organized shelves, proper waste disposal, odor control, and the general neatness of the service environment all influence customers' perceptions of safety. These visible indicators shape their assessment of an establishment's sanitation performance and reflect how effectively food facilities apply sanitation and safety procedures.

Utensils & Equipment Handling

Proper handling and sanitization of utensils and equipment are essential components of safe food preparation because they prevent cross-contamination and reduce the risk of foodborne illness. A study on street food vendors in Calapan City showed that food handlers demonstrated generally good hygiene practices, particularly in cleaning and sanitizing equipment. Tyler et al. (2023) found a strong positive correlation (r = 0.631) between hygiene procedures and food safety levels, demonstrating that consistent cleaning of utensils and equipment contributes directly to safer food handling.

The importance of proper cleaning practices is also reflected in findings from restaurant settings. Research from Iran involving 28 food establishments revealed high contamination levels on food-contact surfaces such as chopping boards and countertops. Fallahizadeh et al. (2025) identified ATP bioluminescence testing as a more effective method than simple visual inspection for detecting hidden contamination. The notable reduction in contamination after washing highlighted the need for thorough equipment sanitation and regular hygiene training for staff.

Local studies also reinforce the value of proper utensil and equipment management. An assessment of canteen food handlers in the Philippines showed moderate knowledge and practices in utensil sanitation, with the most significant gaps found in temperature control and cross-contamination prevention. Maye et al. (2024) noted that training and experience were important factors in improving compliance with food safety protocols. Consistent with these findings, this research gives attention to visible aspects of utensil and equipment handling that customers can directly observe, including the cleanliness of service utensils such as tongs, trays, and scoops, as well as display cases, countertops, and packaging tools. These indicators influence how customers assess the sanitation level of a bakery and the safety of the products being offered.

The reviewed literature shows that food safety and sanitation in bakeries are influenced by a combination of individual, organizational, and environmental factors. Personal characteristics

such as age, educational attainment, and sex affect food safety knowledge, attitudes, and behaviors, with older and more educated individuals generally demonstrating better practices. Organizational factors, including business location, workforce size, and years of operation, shape the ability of establishments to maintain hygienic conditions, provide adequate supervision, and develop consistent routines that support sanitation. Studies consistently report that proper food safety and sanitation practices depend not only on knowledge but also on training, resources, and supportive working conditions. Visible aspects of hygiene, such as personnel cleanliness, facility sanitation, and utensil and equipment handling, strongly influence customer assessment, which plays an important role in shaping trust and perceived safety. Overall, the literature highlights that food safety in bakeries results from the combined effects of employee behavior, organizational capacity, environmental conditions, and customer evaluation, all of which must work together to ensure safe and reliable food products.

1.4 Schematic Diagram

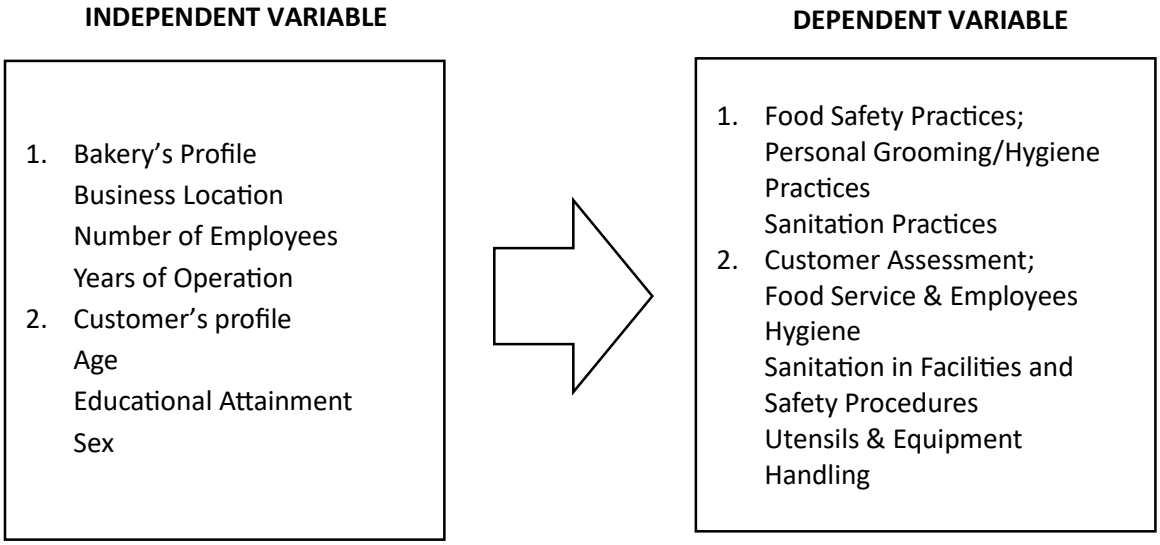


Figure 1 *shows the conceptual flow illustrating how the bakery and consumer profiles influence food safety and sanitation practices.*

1.5 Statement of the Problem

This study aims to assess the food safety and sanitation practices of local bakeries in the municipalities of Buenavista, Nasipit, and Carmen, Agusan del Norte. Specifically, it seeks to answer the following questions: Specifically, the study seeks to answer the following questions:

1. What is the profile of the bakeries in terms of:

   1.1 business location:

   1.2 number of employees; and

   1.3 years of operation?

2. What is the profile of the customers in terms of:

   2.1 age;

   2.2 educational attainment; and

   2.3 sex?

3. What are the food safety and sanitation practices among local bakeries in the municipalities of Buenavista, Nasipit, and Carmen in terms of:

   3.1 personal grooming/hygiene practices; and

   3.2 sanitation practices?

4. How do the customers assess the safety and sanitation practices in selected local bakeries in the municipalities of Buenavista, Nasipit, and Carmen in terms of:

   4.1 food service and personnel's hygiene;

   4.2 sanitation in facilities and safety procedures; and

   4.3 utensils and equipment handling?

5. Is there a significant relationship between the profile of respondents and the food safety and sanitation practices among local bakeries in the municipalities of Buenavista, Nasipit, and Carmen?



1.6 Hypothesis

Null Hypotheses (H_o)

- H_o: There is no significant relationship between the profile of the respondents (business profile and customer profile) and the food safety and sanitation practices among local bakeries in the municipalities of Buenavista, Nasipit, and Carmen, Agusan del Norte.

1.7 Scope and Limitation

This study focuses on assessing the food safety and sanitation practices of registered bakeries in the municipalities of Buenavista, Nasipit, and Carmen, Agusan del Norte, which have been operating for at least five (5) years to ensure stable and consistent operations. Participation is voluntary, and only individuals who agree to take part are included; those who decline are not forced or counted. The study excludes unregistered or newly established bakeries and does not include full demographic profiling of bakery staff, as the focus is on establishment-level factors that influence food safety and sanitation. The study also excludes respondents below 18 years old to ensure that only legally responsible consumers participate. For the bakery respondents, only one representative per establishment is included, specifically the owner, manager, supervisor, or the staff member in charge of food safety or food handling; other employees are not part of the respondent group. The findings reflect practices and consumer perceptions during the data collection period and may not represent other bakeries outside the selected areas.

1.8 Definition of Terms

For clarity and consistency, the following terms are operationally defined as they are used in this study:

Age. This refers to the customer's age group as indicated in the consumer survey (18–29, 30–39, 40–49, or 50 years and above).

Business Location. This refers to whether the bakery is situated in a rural or urban area, as indicated in the bakery profile checklist.

Bakery Respondent. This refers to the owner / manager / supervisor / staff responsible for sanitation during data collection.

Customer Assessment. This refers to customers' evaluation of bakery hygiene and sanitation practices, based on their ratings of food service hygiene, facility sanitation, and utensils/equipment handling in the consumer assessment instrument.

Educational Attainment. This refers to the highest educational level completed by the customer, categorized in the survey as: student, high school graduate, college graduate, postgraduate, or prefer not to say.

Food Safety. This refers to the condition of food being safe for consumption, as described by the bakery's adherence to hygiene and sanitation practices measured in the instruments.

Food Safety Practices. This refers to the actions performed by bakery personnel to maintain cleanliness and prevent contamination, measured through their responses to items under personal hygiene and sanitation practices in the bakery instrument.

Food Service and Personnel Hygiene. This refers to customers' evaluation of staff hygiene and food protection practices specifically bread covering, minimizing bare-hand contact, use of hairnets, aprons, gloves, and visibility of sanitary permits as assessed in the consumer instrument.

Local Bakeries. This refers to selected bakeries located in Buenavista, Nasipit, and Carmen, Agusan del Norte that produce and sell bread and pastries to the local community.

Number of Employees. This refers to the total count of workers employed in the bakery at the time of the study, as reported in the bakery profile survey.

Policy Enhancement. This refers to recommended improvements in bakery sanitation monitoring and food safety guidelines based on study results.

Sanitation in Facilities and Safety Procedures. This refers to customers' evaluation of bakery facility cleanliness and safety conditions, specifically the functionality and cleanliness of lighting, adequacy of space, and availability of garbage containers as measured in the consumer instrument.

Sanitation Practices. This refers to bakery staff's cleaning and sanitation routines—including cleaning tools, sanitizing surfaces, checking sinks, proper use of utensils, proper cooking temperatures, and minimizing bare-hand contact—measured through responses in the bakery sanitation checklist.

Sex. This refers to the customer's biological classification (male or female) as indicated in the consumer survey.

Utensils and Equipment Handling. This refers to customers' assessment of how utensils and equipment are cleaned, maintained, and used specifically the absence of stains/soap, dust-free equipment, and correct utensils used when serving bread.

Years of Operation. This refers to how long a bakery has been running. The length of time in business often reflects experience and how well the bakery has adapted to food safety standards.

CHAPTER 2

RESEARCH METHODOLOGY

This chapter will discuss the research design and description of research respondents, instruments, ethical standards, data gathering procedure, and statistical treatment use.

2.1 Research Design

This study will employ a descriptive–quantitative research design using survey questionnaires to assess food safety and sanitation practices among local bakeries and their consumers. This method is appropriate for obtaining measurable and objective data from both bakery owners and consumers, allowing analysis of their level of compliance and perception toward sanitation practices.

2.2 Research Materials and Methods

Two adapted and refined survey questionnaires will be utilized to gather data from bakery owners and consumers. The bakery questionnaire will be based on the study of (Bihag et. al, 2021), while the consumer questionnaire will be adapted, modified, and updated from the validated instrument of (Picardal et al., 2019) to align with current standards. Both instruments will be revised to fit the bakery context and will undergo content validation and reliability testing to ensure accuracy and appropriateness for this study.

The bakery questionnaire will have two parts. The first part will cover the bakery's demographic profile, including the number of employees, years of operation, and business location. The second part will contain 20 statements on food safety practices, specifically on personal grooming, hygiene, and sanitation. Responses will be rated using a four-point Likert scale*: 4 – Strongly Agree, 3 – Agree, 2 – Disagree, and 1 – Strongly Disagree.*

The consumer questionnaire will also have two parts. The first part will gather demographic information such as age, sex, and educational attainment. The second part will

consist of 11 statements evaluating the consumers' perception of food service and personnel hygiene, sanitation in facilities, safety procedures, and utensils and equipment handling. It will use a five-point Likert scale: *5 – Excellent, 4 – Above Average, 3 – Average, 2 – Below Average, and 1 – Very Poor.* The questionnaire is intended to be answered by the bakery owner; however, if the owner is not available at the time of data collection, the designated person-in-charge such as a manager, supervisor, or staff member directly responsible for food safety and sanitation will be permitted to answer on behalf of the establishment to ensure accurate representation of operational practices.

Before data collection, both instruments will undergo expert validation to ensure clarity, content relevance, and appropriateness. A pilot test will then be conducted to determine their reliability using Cronbach's Alpha. Using separate questionnaires for bakery owners and consumers is expected to provide more accurate and comprehensive data on food safety and sanitation practices in local bakeries.

2.3 Research Environment, Population and Sample

This study will examine the food safety and sanitation practices of registered local bakeries based on the official municipal listings of Buenavista, Nasipit, and Carmen, Agusan del Norte. It will focus on key areas such as personal hygiene, food handling, cleaning and sanitation, storage practices, and waste management to determine how these establishments maintain proper standards in their daily operations.

Research Locale

Buenavista, Nasipit, and Carmen are neighboring municipalities in Agusan del Norte, located within the Caraga Region of the Philippines. These areas were chosen as the study locale because of their growing number of registered bakeries and the consistent consumer demand for

baked products. The study will include only bakeries that are officially registered in these municipalities to ensure that the data collected are consistent, reliable, and verifiable.

Population and Sampling

During the initial data-gathering phase, the researchers visited each bakery in Buenavista and Carmen since the municipal offices could not provide data on their years of operation. The team personally verified each bakery and identified those that had been operating for at least five (5) years. In Nasipit, the municipality provided complete registration data, including the years of operation. Only bakeries that met this five-year qualifier were included in the study to ensure that the establishments have stable operations and established practices. The final count includes 13 bakeries in Buenavista, 11 in Nasipit, and 4 in Carmen, for a total of 28 bakeries.

For the consumer respondents, the researchers inquired at each bakery about their average number of daily customers. These data were used to estimate the number of consumer respondents per location. Consumers will be selected through convenience sampling, meaning those who will be available and willing to participate during the researchers' visits will be included. Consumer respondents must be ≥ 18 years old. Individuals under 18 will be excluded from the consumer sample. A total of 316 consumer respondents was determined using Cochran's formula and will be proportionally distributed based on the number of customers reported by each bakery.

Table 1. Respondents of the study

RESPONDENT	Population	Sample	
	F	f	%
BUENAVISTA			
BAKESHOP 1	50	10	3.16
BAKESHOP 2	12	2	0.63
BAKESHOP 3	150	29	9.18
BAKESHOP 4	20	4	1.27
BAKESHOP 5	20	4	1.27
BAKESHOP 6	50	10	3.16

BAKESHOP 7	40	8	2.53
BAKESHOP 8	50	10	3.16
BAKESHOP 9	10	2	0.63
BAKESHOP 10	30	6	1.90
BAKESHOP 11	50	10	3.16
BAKESHOP 12	290	56	17.72
BAKESHOP 13	25	5	1.58
NASIPIT			
BAKESHOP 1	80	15	4.75
BAKESHOP 2	40	8	2.53
BAKESHOP 3	30	6	1.90
BAKESHOP 4	50	10	3.16
BAKESHOP 5	100	19	6.01
BAKESHOP 6	10	2	0.63
BAKESHOP 7	70	13	4.11
BAKESHOP 8	25	5	1.58
BAKESHOP 9	50	10	3.16
BAKESHOP 10	120	23	7.28
BAKESHOP 11	30	6	1.90
CARMEN			
BAKESHOP 1	10	2	0.63
BAKESHOP 2	60	12	3.80
BAKESHOP 3	100	19	6.01
BAKESHOP 4	50	10	3.16
TOTAL 28	**1622**	**316**	**100.00**

Table 1 *Presents the Distribution of Bakery and Consumer Respondents. All 28 bakeries that met the inclusion criteria (registered and operating for at least five years) will be included through purposive sampling, with one questionnaire per establishment. For consumers, 316 respondents will be surveyed, determined using Cochran's formula and proportionally distributed based on each bakery's average number of daily customers.*

2.4 Research Data and Analysis Plan

The researchers will use purposive sampling for bakery respondents and convenience sampling for consumer respondents. This method is appropriate because it allows the inclusion of bakeries that meet specific criteria (i.e., registered and operating for at least five years) while ensuring accessibility to willing consumers.

Statistical Treatment

In presenting, analyzing, and interpreting the data, the following statistical tools will be used:

- **Frequency Count and Percentage** – to answer SOP 1 and SOP 2, determining the profile of bakery and consumer respondents.
- **Weighted Mean and Standard Deviation** – to answer SOP 3 and SOP 4 assessing the level of food safety and sanitation practices among local bakeries and the consumers' evaluations.
- **Pearson's Correlation Coefficient** – to determine the relationship between the respondents' profiles and their assessment of food safety and sanitation practices. This test is appropriate because the study examines the relationship between quantitative variables, such as age, years of operation, and hygiene or sanitation ratings, allowing the researchers to identify whether significant correlations exist among these variables.

2.5 Research Ethical Standards

A. **Intellectual Property Rights**

The study will respect all intellectual property rights by ensuring originality and proper acknowledgment of sources. All secondary data, references, and adapted research instruments will be appropriately cited to honor the intellectual property of other researchers. Respondent anonymity will be maintained throughout the data collection and analysis process to protect both participants and the researchers' own intellectual contributions.

B. Informed Consent

The respondents will be bakery staff and consumers from selected local bakeries in Buenavista, Nasipit, and Carmen (BueNasCar). Before data collection, the researchers will conduct a brief orientation to explain the study's purpose, objectives, and potential benefits and risks. Participants will be informed of their right to refuse or withdraw from participation at any time

without any consequences. Written informed consent forms will be distributed to ensure that all respondents clearly understand their rights and voluntarily agree to participate.

C. Data Privacy and Confidentiality

All personal data to be collected, such as demographic information and responses, will be treated with strict confidentiality. Respondent identities will be anonymized and coded to protect their privacy. Only the researchers will have access to the dataset and the identification codes. Data will be securely stored and used solely for academic purposes. The confidentiality of all participants will be upheld throughout the research process.

D. Voluntary participation

Participation in the study will be voluntary. Respondents will be informed of their right to withdraw at any time without any negative consequences. If a participant decides to withdraw, all of their data will be excluded from the study and securely deleted to protect their confidentiality. No animals will be involved in this study; therefore, considerations for alternatives to animal use are not applicable.

E. Minimization of Harm

The researchers will ensure a safe, respectful, and professional environment throughout data gathering. The study will pose minimal risk to participants. Should any discomfort or concern arise, respondents will be referred to appropriate support services such as the Municipal Health Office or the Guidance Office of Saint Michael College of Caraga. A debriefing will be provided after participation to ensure that no emotional, psychological, or physical harm occurs.

F. Beneficence

The study will benefit the community by identifying gaps and challenges in food safety and sanitation practices among local bakeries in BueNasCar. The findings may serve as a guide for policymakers, business owners, and consumers to improve food handling practices and ensure

consumer safety. Respondents will be allowed to request access to the results upon completion of the study.

G. Justice and Fair Participant Selection

Respondents will be selected based on well-defined and fair inclusion criteria, ensuring that the selection process aligns with the study's objectives. Respondents from diverse backgrounds will be included fairly without discrimination.

H. Data Integrity and Accuracy

Respondents will be selected based on fair inclusion criteria aligned with the research objectives. Only officially registered bakeries listed in municipal records and has been operating for at least 5 years and consumers who regularly purchase bakery products will be included. The selection will be done without discrimination based on sex, age, educational background, or social status, ensuring fairness and equity.

I. Transparency in Reporting

The study findings will be reported honestly, with no manipulation or alteration of data. All sources, such as academic references, will be cited appropriately and acknowledged to ensure respect for intellectual property rights and uphold ethical research standards. The study will utilize AI tools to support various aspects of the research. There will be no potential conflict of interest involved in the study.

J. Use of Patented or Copyrighted Materials

No patented materials will be involved in this study. All copyrighted materials, including references, will be appropriately cited in accordance with intellectual property laws. The researchers will ensure that all sources are appropriately acknowledged, maintaining academic integrity and avoiding plagiarism.

K. Considerations for Human Subjects

The researchers will inform the respondents about the study's possible risks and potential benefits, ensuring that an informed consent form will be provided. If any significant risks arise, such as emotional, physical, or psychological concerns, appropriate health monitoring programs and emotional support services will be made available.

L. Responsible Use of AI and Related Technology

The researchers will utilize digital technologies throughout the research process, including sourcing reliable articles to maintain the accuracy and credibility of the study. AI tools, such as ChatGPT, will assist in summarizing statements and enhancing the overall quality of the paper. Platforms like Google Scholar, Elicit, and Typeset.io will be employed to locate supporting studies, ensuring the reliability and credibility of the research. Additionally, Grammarly will be used to ensure that the research report adheres to proper grammar and clarity standards. The responsible use of these technologies will help maintain the integrity of the study.

M. Ethical Approval

The research will obtain ethical approval from the Institutional Review Board of Saint Michael College of Caraga to confirm compliance with all relevant guidelines on respondent protection, data privacy, and intellectual property rights.

2.6 Data Gathering Procedure

In gathering the data, the researchers will first secure endorsement letters addressed to the municipal offices of Buenavista, Nasipit, and Carmen to formally request the official list of registered bakeries in each municipality. Once the lists are obtained, the researchers will prepare and finalize the survey questionnaires for both bakery owners/employees and consumers. The instruments will undergo expert validation to ensure clarity, relevance, and accuracy, followed by a pilot test to determine reliability using Cronbach’s Alpha. Upon approval, the survey will be administered to all listed bakeries using purposive sampling for bakery respondents, while

convenience sampling will be employed for consumers, targeting customers who are available and willing to participate at the selected bakery locations.

Only customers aged 18 years and above will be included in the study; their age will be verified at the time of survey administration to ensure compliance with this criterion. Data collection will be conducted on weekends (Saturday and Sunday) to capture regular bakery operations and customer flow during non-working days. Surveys will be administered in two sessions: 6:00–9:00 AM for morning customers and 3:00–5:00 PM for afternoon customers. Each respondent is expected to spend approximately 30 minutes completing the survey. For bakery respondents, the survey will be conducted on-site at their establishments, while consumers will be approached during the designated periods to ensure voluntary participation. After data collection, all responses will be tallied, encoded, and submitted to the statistician for analysis and interpretation, ensuring that the study accurately reflects bakery food safety practices and consumer assessment in the selected municipalities.

A. BAKERIES PROFILE

Direction. Please read each item carefully and place a check (✓) in the column that best represents your actual practice. Your responses are confidential and will only be used for research purposes.
There are no right or wrong answers. Answer honestly based on your usual work routines.
(Palihog basaha ang matag-item ug ibutang ang tsek (✓) sa kolum nga nagpakita sa imong tinuod nga binuhatan. Ang imong tubag kay kompidensyal ug alang lamang sa research. Wala'y sakto o sayop nga tubag. Tubaga tinuoray sumala sa imong kasagarang praktis.)

Business Location

☐ Rural

☐ Urban

Number of Employees:

☐ 1–3

☐ 4–6

☐ 7–10

☐ 11 and above

Years of Operation

☐ 5–10 years

☐ 11–15 years

☐ 16–20 years

☐ 21 years and above

Checklist Scale	Verbal Description	Cebuano Translation	Verbal Interpretation
4	**Strongly Agree**	*(Labing Uyon)*	I/We consistently do this / Always practice this behavior.
3	**Agree**	*(Uyon)*	I/We often do this / Frequently practice this behavior.
2	**Disagree**	*(Dili Uyon)*	I/We seldom do this / Occasionally practice this behavior.
1	**Strongly Disagree**	*(Labing Dili Uyon)*	I/We never do this / Rarely or never practice this behavior.

B. FOOD SAFETY PRACTICES

1. PERSONAL HYGIENE/GROOMING PRACTICES

Statements/Questions	**Strongly Agree** (Labing Uyon) **4**	**Agree** (Uyon) **3**	**Disagree** (Dili Uyon) **2**	**Strongly Disagree** (Labing dili Uyon) **1**

1. Check the self-health condition and body temperature every working day. (Pagsusi sa kahimtang sa kahimsog sa kaugalingon ug temperatura o kainiton sa lawas matag adlaw sa pagtrabaho.)				
2. Checking cleanliness of clothes, hair restraints, and prescribed shoes before working. (Pagsusi sa kalimpyo sa sanina, pagpugong sa buhok, ug pagsuot ug sapatos ayha magsugod sa magtrabaho.)				
3. Washing hands before handling foods. (Pagpang-hugas sa mga kamot sa dili pa muhikap sa mga pagkaon.)				
4. Ensuring fingernails are kept short, clean, and without nail polish to avoid possible food contamination. (Pagpahimutang nga ang mga kuko mubo, limpyo, ug walay kolor aron malikayan ang posibleng pagkontamina sa pagkaon.)				
5. Wearing gloves is necessary when handling food, especially for those who have abrasions or cuts on their hands. (Gikinahanglan ang pagsuot ug gwantes kung muhikap sa mga pagkaon labina kung adunay samad sa ilang kamot.)				
6. Wearing accessories like rings, watches, and bracelets is prohibited when handling foods. (Ginadili ang pagsul-ob ug mga aksesorya sama sa mga singsing, relo, ug polseras kung maghimo sa pagkaon.)				
7. Avoid eating in the kitchen while preparing foods. (Ginadili and pagkaon sa kusina samtang nag andam sa mga pagkaon.)				
8. Wearing a mask while preparing food is an important practice to reduce the risk of food contamination. (Pagsuot ug maskara kung nag-andam ug pagkaon maoy usa ka importante nga buluhaton aron malikay ang pagkaon sa kontaminasyon.)				
9. Wearing PPE is an important practice to prevent the risks in the workplace and to avoid food contamination. (Ang pagsuot sa saktong sapot pangkusina maoy importanteng buluhaton aron malikayan ang disgrasya sa gi-trabahoan ug masugpo ang kontaminasyon sa pagkaon.)				
10. Avoid always talking with coworkers while preparing food.				

(Ginadili ang pag sige'g estorya sa kauban kung naghikay sa pagkaon.)				

2. SANITATION PRACTICES

Statement/Questions	**Strongly Agree** (Labing Uyon) **4**	**Agree** (Uyon) **3**	**Disagree** (Dili Uyon) **2**	**Strongly Disagree** (Labing Dili Uyon) **1**
1. Always clean and sanitize the kitchen tools and equipment and store them properly using a shelving unit. (Kanunayon ang paglimpyo ug sanitisa sa mga gamit sa kusina og iplastar sa insaktong butanganan.)				
2. Always clean and sanitize work areas, cutting boards, knives, and utensils. (Kanunayon ang paglimpyo ug sanitisa sa lugar nga gitrabahuan, taparan, kutsilyo og uban pang mga gamit.)				
3. Always use the serving utensils with long handles when stirring large amounts of food. (Kanunayon ang paggamit og luwag nga nay taas nga haweranan og magkutaw og dinaghan nga niluto nga pagkaon.)				
4. Always use gloves or utensils to handle food after cooking. (Kanunayon ang paggamit og gloves kung mag-alsa o maghawid og pagkaon nga humana na gi luto.)				
Always cook the food properly within the required time at the required temperature. (Kanunayon ang pagluto sa pagkaon sa sulod sa sakto og girekomenda nga oras og temperature.)				
Minimizing bare-hands contact with foods. (Minusa o likayan ang paghikap sa pagkaon labi na kung walay gloves.)				
Always check the availability of hand sanitizer. (Kanunayon ang pagsuri kung aduna ba'y saktong supply sa hand sanitizer.)				
Always clean and check the sink for washing utensils. (Kanunayon ang pagsuri og paglimpyo sa lababo alang sa paghugas sa mga gamit.)				

Avoid smoking in the kitchen while preparing food. (Ginadili ang pagpanigarilyo sa kusina samtang nag andam ug pagkaon.)				
Always clean and store the kitchen tools and equipment properly. (Kanunayon ang paglimpyo og insaktong pagplastar sa mga gamit pang kusina.)				

B. CUSTOMER PROFILE

Direction. Please check (✓) in the required information in the box provided.
(Palihug sa pag tsek (✓) sa gikinihanglan nga kasayuran sa kahon nga gihatag.)

Name (optional): _____

A. Age:
- ☐ 18–29 years old
- ☐ 30–39 years old
- ☐ 40–49 years old
- ☐ 50 years old and above

B. Educational Attainment:
- ☐ Currently studying (Student)
- ☐ High School Graduate
- ☐ College Graduate
- ☐ Postgraduate (Master's/Doctorate)
- ☐ Prefer not to say

C. Sex:
- ☐ Female
- ☐ Male

Directions: Put a check (✓) on the number that corresponds to your answer based on how well you think the bakery and its personnel perform the following safety and sanitation practices. Please rate each item as honestly as possible.

(Ibutang ang tsek (✓) sa numero nga angay sa imong tubag base sa imong pagtan-aw kung unsa ka maayo ang panaderya ug mga trabahante niini sa mga mosunod nga praktis sa kahinlo ug kaluwasan. Palihog pagtubag sa tinuod.)

Checklist Scale	Verbal Description	Verbal Interpretation	Cebuano Translation
5	Excellent	Outstanding / Always meets this standard	Hilabihan ka Maayo / Kanunay makatuman sa sumbanan
4	Above Average	Better than expected / Often meets this standard	Labaw sa Kasagarang Maayo / Kasagaran makatuman sa sumbanan
3	Average	Meets expectations / Sometimes meets this standard	Kasagaran Maayo / Usahay makatuman sa sumbanan
2	Below Average	Less than expected / Seldom meets this standard	Ubos sa Kasagaran Maayo / Dili kasagaran makatuman sa sumbanan
1	Very Poor	Rarely or never meets this standard	Hilabihan ka Dili Maayo / Halos dili o dili gyud makatuman sa sumbanan

1. Food Service and Employees' Hygiene

Statement/Questions	**Excellent** *(Hilabihan ka Maayo)*	**Above Average** *(Labaw sa Kasagarang Maayo)* **4**	**Average** (Kasagarang Maayo) **3**	**Below Average** *(Ubos sa Kasagarang Maayo)* **2**	**Very Poor** *(Hilabihan ka Dili Maayo)*

	5				**1**
1. The bakery ensures that all bread is covered against all sorts of contaminants. (Ang panaderya nagsiguro nga ang tanan nga pan natabonan sa tanan nga mga matang sa mga kontaminante.)					
2. The bakery personnel minimize bare-hand contact with bread. (Ang nag-andam og pagkaon nag-minos sa paggamit ug walay putos nga kamot kung mo gunit sa pan.)					
3. The personnel is/are wearing hairnets and other hair restraints. (Ang mga personnel nagsul-ob ug hairnets ug ubang mga gamit para pagpugong sa buhok.)					
4. The personnel is/are wearing aprons and disposable gloves. (Ang mga personnel nagsul-ob ug apron, ug disposable nga gloves.)					
5. The bakery has a sanitary permit that is easily seen by the customers. (Ang panaderya adunay sanitary permit nga sayon ra makita sa mga kustomer.)					

2. Sanitation in Facilities and Safety Practices

Statement/Questions	**Excellent** *(Hilabihan ka Maayo)*	**Above Average**	**Average** (Kasagarang Maayo)	**Below Average**	**Very Poor**

	5	*(Labaw sa Kasagarang Maayo)* **4**	**3**	*(Ubos sa Kasagarang Maayo)* **2**	*(Hilabihan ka Dili Maayo)* **1**
1. The bakery's lightings are all working and free from dust. (Ang mga suga sa panaderya kay nag-function tanan ug limpyo gikan sa abog.)					
2. The bakery has sufficient space for personnel and customers. (Ang panaderya adunay igo nga luna para sa mga personnel ug mga kustomer.)					
3. The bakery has garbage containers. (Ang panaderya adunay mga basurahan.)					

3. Utensils and Equipment Handling

Statement/Questions	**Excellent** *(Hilabihan ka Maayo)* **5**	**Above Average** *(Labaw sa Kasagarang Maayo)* **4**	**Average** (Kasagarang Maayo) **3**	**Below Average** *(Ubos sa Kasagarang Maayo)* **2**	**Very Poor** *(Hilabihan ka Dili Maayo)* **1**
1. The utensils of the bakery are free from all traces of stains and soap/detergent. (Ang mga gamit sa panaderya limpyo ug walay hugaw, mantsa, o sabon/detergent.)					
2. The equipment that the bakery uses is/are properly cleaned and free from dust. (Ang mga kagamitan nga gigamit sa					

panaderya kay limpyo ug walay abog.)					
3. The bakery uses proper utensils while serving bread. (Ang panaderya naggamit ug husto nga mga gamit sa pagserbisyo sa tinapay.)					

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Personal Information

Sex: FEMALE

Date of birth: AUGUST 11, 2005

Height: 4'11

Weight: 39 kg

Civil Status: SINGLE

Educational Attainment

Elementary: SANTA ANA ELEMENTARY SCHOOL

Secondary: SAINT MICHAEL COLLEGE OF CARAGA

Tertiary: SAINT MICHAEL COLLEGE OF CARAGA

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
LOCAL SUPREME GOVERNMENT	VICE PRESIDENT
SLICE	LADY OF THE HOUSE
SLICE	HEAD OF COMMUNICATION

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Personal Information

Sex: FEMALE

Date of birth: MAY 6, 2004

Height: 5'0

Weight: 50kg

Civil Status: SINGLE

Educational Attainment

Elementary: CARMEN CENTRAL ELEMENTARY SCHOOL

Secondary: OUR LADY OF CARMEN ACADAMY OF CARAGA INCORPORATED

Tertiary: SAINT MICHAEL COLLEGE OF CARAGA

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
SLICE	MEMBER

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Personal Information

Sex: MALE

Date of birth: JANUARY 9, 2005

Height: 5’9

Weight: 73 kg

Civil Status: SINGLE

Educational Attainment

Elementary: MALAYBALAY CITY CENTRAL SCHOOL

Secondary: BUKIDNON NATIONAL HIGH SCHOOL

Tertiary: SAINT MICHAEL COLLEGE OF CARAGA

Membership/Affiliations

ORGANIZATION	POSITIONS HELD
LOCAL SUPREME GOVERNMENT	BUSINESS MANAGER
SLICE	MEMBER

CURRICULUM VITAE



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Personal Information

Sex: FEMALE

Date of birth: FEBRARY 18, 2003

Height: 4’9

Weight: 45kg

Civil Status: SINGLE

Educational Attainment

Elementary: VALENCIA ELEMENTARY SCHOOL

Secondary: SENIOR HIGH SCHOOL CARMEN

Tertiary: SAINT MICHAEL COLLEGE OF CARAGA

Membership/Affiliations

ORGANIZATION	**POSITIONS HELD**
SLICE	MEMBER

CURRICULUM VITAE



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Personal Information

Sex: FEMALE

Date of birth: JULY 12, 2005

Height: 4'11

Weight: 52kg

Civil Status: SINGLE

Educational Attainment

Elementary: NASIPIT CENTRAL ELEMENTARY SCHOOL

Secondary: SAINT MICHAEL COLLEGE OF CARAGA

Tertiary: SAINT MICHAEL COLLEGE OF CARAGA

Membership/Affiliations

ORGANIZATION	**POSITIONS HELD**
SLICE	MEMBER

CURRICULUM VITAE



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Personal Information

Sex: MALE

Date of birth: SEPTEMBER 25, 2004

Height: 5’8

Weight: 80kg

Civil Status: SINGLE

Educational Attainment

Elementary: TAPNIGUE ELEMENTARY SCHOOL

Secondary: SANGAY NATIONAL HIGH SCHOOL

Tertiary: SAINT MICHAEL COLLEGE OF CARAGA

Membership/Affiliations

ORGANIZATION	**POSITIONS HELD**
SLICE	MEMBER